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Q1 - Pruning Calculation in 8-City TSP

Title: Pruning in 8-City Traveling Salesperson Problem (TSP)

- **Problem Context:**
 - Search tree contains 42 nodes at depth level 2.
 - Initial Upper Bound (Greedy Solution): Tour length = 10.

- **Pruning Logic:**

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Condition: Terminate a node if its **Lower Bound (LB)** > 10.

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LB Calculation: $LB = \text{Current Path Length} + \text{Distance to Farthest City} + \text{Distance from Farthest to Start}$.

- **Implementation:**

- At Depth 2, two cities are visited from Start (City 0).
- Examine all 42 combinations (e.g., $0 \rightarrow 1 \rightarrow 2$).
- Paths straying too far (e.g., towards City 3 or 7) often exceed the threshold of 10.

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Goal: Identify and prune suboptimal paths early to significantly reduce tree size.

Q2 - Lower Bound Estimation in Flow Shop Scheduling

Title: Lower Bound Specification for m-Machine n-Job Flow Shop

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Objective: Minimize the maximum completion time (**Makespan**).

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State: Each node represents a partial schedule where k jobs are sequenced.

- **Lower Bound Estimation Strategy:**

1. **Scheduled Time:** Exact completion time for the k sequenced jobs on each machine.

- 2.

Unscheduled Processing: Sum of minimum processing times for the remaining $n - k$ jobs on a bottleneck machine.

3. **Tail Time:** Minimum theoretical time to complete remaining processes on subsequent machines after the bottleneck.

- **Decision:** If $LB > \text{Current Best Makespan}$, terminate the search at this node.

Slide 3: Q3 - Practical Limits of Branch and Bound (B&B)

Title: Computational Scalability of B&B for 10-Machine Problems

- **Theoretical Complexity:**
 - An n -job problem has $n!$ possible solutions.
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 - $n = 14$: ~87 billion solutions (Manageable with good algorithms).
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 - $n = 100$: Unrealistically large search space.

- **B&B Efficiency:**

- Reduces computation by pruning nodes that cannot yield optimal results.
- However, it cannot fully overcome the exponential growth of NP-hard problems.

- **Estimated Limits:**

- For $m = 10$, absolute optimal solutions are typically achievable for $n \approx 20\text{--}30$ within a day.
- For larger n (e.g., 100), metaheuristics (Genetic Algorithms, SA, TS) must be used to find near-optimal solutions.